

Práctica 5.1. Diseño de controlador para un sistema de segundo orden

Diseño de un controlador para un sistema de segundo orden

• Ecuaciones principales

$$V_e(t) = Ri_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)]$$

$$L = \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = Ri_2(t) + Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_e(s) = RI_1(s) + Ls[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)]$$

$$Ls[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)] = RI_2(s) + \frac{I_2(s)}{Cs}$$

$$V_s(s) = RI_2(s) + \frac{I_2(s)}{Cs}$$

Procedimiento algebraico

$$V_e(s) = (Ls + 2R)I_1(s) - (Ls + R)I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R)I_2(s) = \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s)$$

$$V_s(s) = (R + \frac{1}{Cs})I_2(s) = \frac{CRs + 1}{Cs} I_2(s)$$

$$V_e(s) = \left[(Ls + 2R) \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} - (Ls + R) \right] I_2(s)$$

$$= \frac{3CLR s^2 + (5(R^2 + L)s + 2R)}{CLs^2 + CRs} I_2(s)$$

Función de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{[polinomio] I_2(s)}{[?s^2 + ?s + ?] I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CR_s + 1}{C_s} I_2(s)}{\frac{3CLR_s^2 + (5CR^2 + L)s + 2R}{CL_s^2 + CR_s} I_2(s)}$$

$$= \frac{(CR_s + 1)(L_s + R)}{3CLR_s^2 + (5CR^2 + L)s + 2R}$$

Error en estado estacionario

$$\frac{V_s(s)}{V_e(s)} = \frac{CLR_s^2 + (CR^2 + L)s + R}{3CLR_s^2 + (5CR^2 + L)s + 2R}$$

$$e(s) \lim_{s \rightarrow 0} \frac{1}{s} \left(1 - \frac{CLR_s^2 + (CR^2 + L)s + R}{3CLR_s^2 + (5CR^2 + L)s + 2R} \right)$$

$$\lim_{s \rightarrow 0} 1 - \frac{1}{2}$$

- Error en estado estacionario
- Estabilidad en lazo abierto

$$e(s) = \frac{1}{2}$$

$$0.5$$

$$R = 4.7 \text{ k}\Omega$$

$$L = 10 \text{ mH}$$

$$C = 330 \text{ nF}$$

$$R = 4.7 \times 10^3 \Omega$$

$$L = 10 \times 10^{-3} \text{ H}$$

$$C = 330 \times 10^{-6} \text{ F}$$

$$a = 3CLR = 0.04653$$

$$b = 5CR^2 + L = 36448.51$$

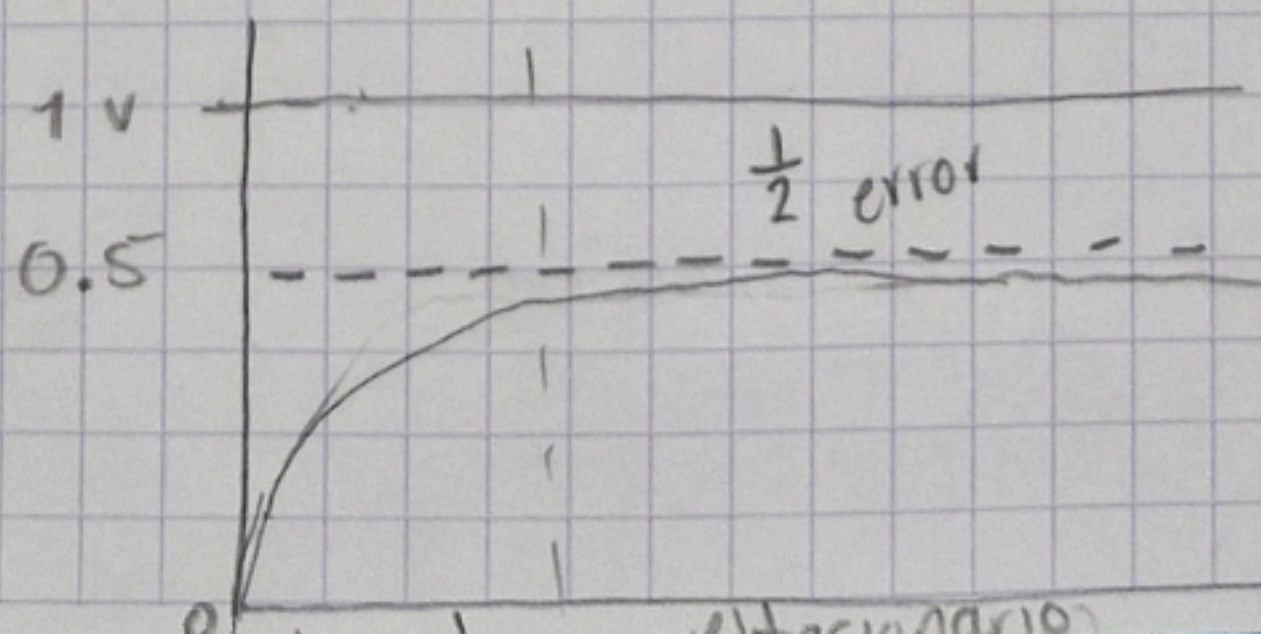
$$c = 2R = 9400$$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda_1 = \frac{-5CR^2 + L + \sqrt{(5CR^2 + L)^2 - 4(3CLR)(2R)}}{2(3CLR)} = -0.2578981442$$

$$\lambda_2 = \frac{-5CR^2 + L - \sqrt{(5CR^2 + L)^2 - 4(3CLR)(2R)}}{2(3CLR)} = -783333.2904$$

Números negativos reales y diferentes, el sistema tiene respuesta sobreamortiguada.



$$3CLR = 0.000047$$

$$5CR^2 + L = 36.458500$$

$$2R = 9400$$

Modelo de Ecuaciones Integro-diferenciales

El modelo se expresa despejando las variables dependientes, es decir, las corrientes.

$$V_e(t) = 2Ri_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - Ri_2(t)$$

$$i_1(t) = \left[V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$L \cdot \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) = Ri_2(t) + Ri_2(t) + Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$